

General Course Information

Course name: EO (Earth Observation) system

Course ID: MA613

Course Structure: 3 – 0 – 2 - 4

Course Type: Core course, Semester I for M.Sc. Agriculture Analytics

Credit : 4

Contact hours: 5 hours per week

Course Objective: Studying the earth surfaces can be possible using conventional means like traversing or ground survey. Although it is a sound technique for studying the details with the help of various ground- based instrumentation but studying the regional relationship of the earth features and the various processes, past or present, is not possible without the use of appropriate technology. It is in this context the technologies like satellite based remote sensing; Geographic Information System (GIS) are relevant.

Detailed content:

Theory

Sl. No.	Description	No. of
1	Introduction to remote sensing and its applications	1
2	Electromagnetic radiation, Energy matter interaction – refraction, scattering, absorption, reflection, fundamentals of radiometry	2
3	Spectral signature of earth surface features – vegetation, soil, snow, urban etc.	2
4	Platforms for data collection – ground based, aerial and space based. Satellite orbits – geosynchronous and sun synchronous.	2
5	Sensor characteristics – concept of spatial, spectral, radiometric and temporal resolutions	2
6	Space based sensors – multispectral, hyper-spectral, microwave	2
7	Radiometric correction – shot noise, line and pixel drop out, line start correction, striping	1
8	Atmospheric corrections – absolute and relative atmospheric corrections	1
9	Geometric corrections – satellite attitude corrections, corrections for slope and aspect	1
10	Introduction of GIS, datum and coordinate system	2
11	Geographical data types, GIS data model, Map projection	2
12	GIS database organization - Geographic Data Formats, Topology and	3
13	Spatial data analysis with GIS - GIS Operational Procedure, Overlay	3
14	Map accuracy and GIS errors	2
15	Image enhancement – linear and non-linear contrast enhancement, band ratio, neighbourhood raster operations, spatial filtering, principal component analysis, vegetation indices, band ratio, neighbourhood raster operations, spatial filtering, principal component analysis,	2
16	Pattern recognition and Image classification– supervised and unsupervised approach	3
17	Machine learning and neural network in remote sensing	2
18	Introduction to SAR (Synthetic Aperture radar)	2

Evaluation criteria:

Evaluation is based on

- First / Mid and final semester exams. (60%)
- Quizzes in the class (10%)
- Mini project (30 %)

Learning outcomes:

After successful completion of the course, a student

- Explain the physical principles and concepts underlying EO
- Describe the spectral signature of natural objects on earth surface.
- Describe the sources, nature and common form of EO data
- Will be able to understand the technical information about orbit, sensors
- Be aware of new developments and trends in EO

Learning and teaching activity:

- 3 x 1-hour lectures per week
- 1 x 2-hour lab session per week

Suggested text book

1. Remote Sensing Digital Image Analysis: John A. Richards, Xiuping Jia,
2. Introductory Digital Image Processing: John R Jensen